Evaluating Resource Adequacy Using Hourly 40 Weather-Year Data Under High VRE Penetration for the Luzon, Visayas, and Mindanao Grids in Alignment with the Power Development Plan (2023-2050)

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Abstract—**The Philippines is pursuing a major energy transition, targeting a renewable energy share of 35% by 2030 and 50% by 2040, as detailed in the Power Development Plan (PDP) 2023–2050. This rapid integration of variable renewable energy (VRE) sources presents significant challenges to long-term resource adequacy. The PDP's current planning methodology for the Philippines does not account for how unpredictable weather patterns can behave over long periods; specifically, how wind and solar output can drop significantly for extended times. This omission may lead to a systematic underestimation of the capacity needed to ensure supply adequacy during rare but high-impact challenging weather years. This research adapts the net load profiling from Ullmark et al. to the Philippine grid. Using 40 years (1985–2024) of hourly meteorological data from the MERRA-2 reanalysis dataset, multi-decade net load recurrence matrices were generated to evaluate the Department of Energy's expansion scenarios for 2025, 2030, 2040, and 2050. Ultimately, this analysis assesses the sufficiency of planned capacities against the inherent intermittency of VRE resources. Results indicate that as VRE penetration increases, the Luzon and Mindanao grids will face escalating vulnerabilities, transitioning from thin reserve margins in the near term to multi-day capacity shortfalls by 2040.**

Keywords—Resource Adequacy Assessment, Power Development Plan, Energy Planning, Variable Renewable Energy, Weather-years, Recurrence Matrices

I. Introduction

Resource adequacy is a fundamental concept in power system planning that refers to the ability of supply-side and demand-side resources to meet the aggregate electrical demand, including system losses, at all times [1]. This assessment is crucial for ensuring that an electricity system maintains reliable service to consumers while balancing the economic considerations of capacity investments [2]. For systems with substantial variable renewable energy (VRE) penetration, adequacy assessments increasingly rely on net load analysis. In power systems, net load is defined as the total electricity demand minus the power supplied by non-dispatchable energy sources [3]. Net load analysis is important to capture the variability of renewable energy, for power system planning [4]. Sustained periods of high net load, when demand is high while solar and wind output are low, reveal when dispatchable generation and storage must carry the system. To characterize the full range of these events across multiple decades, recent work has used net-load recurrence matrices, also called net-load fingerprints. These matrices count how often a power deficit of a given magnitude persists for a given duration across decades of hourly weather data [5]. These fingerprints capture both peak capacity needs and sustained energy needs in a single picture, making them well-suited for systems dominated by weather-dependent generation.

The Philippine Power Development Plan (PDP) 2023-2050 has established ambitious renewable energy targets, aiming for a 35% share in the power generation mix by 2030 and 50% by 2040 [6]. Achieving these targets requires substantial solar and wind buildout across the Luzon, Visayas, and Mindanao grids, alongside corresponding dispatchable and storage resources.

The integration of VRE sources, particularly solar and wind power, introduces unprecedented challenges to maintaining resource adequacy [7]. Unlike conventional thermal generation, VRE output is inherently dependent on weather conditions, creating variability across multiple timescales, from minute-to-minute fluctuations to seasonal and interannual variations [8]. The net load must be continuously balanced by dispatchable resources such as thermal power plants, hydroelectric facilities, and energy storage systems [5]. As VRE penetration increases, the characteristics of net load variability fundamentally change, potentially creating extended periods where dispatchable capacity is insufficient to meet demand [9].

Existing energy system studies for the Philippines have focused largely on generation expansion planning and low-carbon policy assessment, while the PDP itself uses hourly profiles within a least-cost capacity expansion framework but does not explicitly stress-test system performance against historically challenging weather years [6]. International research has shown that capacity expansion models built on a single weather year or a small sample of years can systematically misrepresent investment requirements for generation and storage [5, 10], and that probabilistic adequacy methods that lack the temporal depth to span multiple decades can underrepresent rare but high-impact weather events [5, 11]. Reference [5] addressed this directly by introducing a net-load fingerprinting methodology that uses 39 years of hourly data to characterize the full range of net load variability for the North European electricity system [5]. Their analysis demonstrated that interannual variability in wind and solar output is large enough that single-year planning systematically understates the peak thermal capacity and long-duration storage needed during the most challenging historical years, and that a small set of carefully selected representative weather years can reproduce the full multi-decadal variability at a fraction of the computational cost.

This study adapts the net-load fingerprinting methodology of [5] to the Philippine context. The analysis uses 40 years of hourly meteorological data spanning 1985 to 2024 from the MERRA-2 reanalysis dataset [12], accessed through Renewables.ninja [13, 14, 15]. From this data, multi-decade net load recurrence matrices are constructed for the Luzon, Visayas, and Mindanao grids under the Reference Scenario and Clean Energy Scenario 1 (CES 1), at five-year intervals from 2025 to 2050. Each recurrence matrix is a heatmap that shows how often the system faced a power shortfall of a given size that lasted at least a given duration. By overlaying the PDP's planned dependable capacities and battery capacities onto these heatmaps, the study evaluates whether the planned fleet can handle the variability of solar and wind output across the full 40-year weather record.

The specific objectives are to generate 40-year hourly solar and wind profiles for each grid, aggregated across the Competitive Renewable Energy Zones (CREZ) [16, 17]; build recurrence matrices that characterize net load variability for each PDP planning year; evaluate the sufficiency of the PDP's planned dependable capacity and four-hour battery storage [18] against these matrices; and identify the capacity-risk and energy-risk zones that show when the planned fleet falls short. The analysis treats each grid as a single node, does not model future climate change, and assumes that future capacity is distributed across the renewable energy zones in proportion to their technical potential.

II. METHODOLOGY

The methodology consists of four parts: (i) data gathering and processing; (ii) model building; (iii) validation; and (iv) simulation and analysis. The Data Gathering and Processing stage focuses on sourcing the meteorological, demand, and generation capacity datasets and converting them into the composite hourly profiles that feed the net load calculation. The Model Building and Validation stages discusses the multi-decadal net load profiling, the construction of the recurrence matrix, and the validation of the model. Finally, the Simulation and Analysis stage tests the planned dependable capacities against the net load profile to identify any capacity gap in the PDP 2023-2050 targets.

A. Data Gathering and Processing

1. *Weather years:* Forty years of hourly meteorological data are sourced from the MERRA-2 reanalysis dataset via the Renewables.ninja platform [13]. The 40-year window provides the temporal depth required to capture interannual variability and rare but high-impact weather events. Rather than manually deriving environmental profiles from raw weather variables, this study uses the Virtual Wind Farm and Virtual PV algorithms of Renewables.ninja, which already account for temperature-dependent efficiency losses, wind speed hub-height extrapolation, and other complex variable interactions.

2. *Load year:* The baseline 8760-hour demand series for the year 2025 across the Luzon, Visayas, and Mindanao grids was obtained from the National Grid Corporation of the Philippines (NGCP) [19]. To establish the demand shape for the entire target year planning, this 2025 profile was normalized where each hourly demand value was divided by the 2025 annual peak demand to produce a per-hour scaling factor. The hourly demand profiles for the target years were subsequently generated by multiplying this normalized baseline shape by the projected annual peak demand for each respective year.

3. *Site selection:* Solar and wind generation were simulated at the geographic centroid of each Competitive Renewable Energy Zone (CREZ) defined in [20]. The total count of CREZ and Offshore Wind (OSW) sites per grid are as follows: Luzon: 13 CREZ and 2 OSW, Visayas: 6 CREZ and 2 OSW, and Mindanao: 4 CREZ. They are all treated as grid-wide profiles constructed as a capacity-weighted spatial average. The specific centroid coordinates are provided in Appendix A.

   To standardize the simulations across all zones, a 1 kW reference system was established for each CREZ using the following baseline parameters:

   - **Solar PV:** Modeled with a 10% system loss, a 10° tilt angle, and an azimuth of 180° (south-facing) [14].
   - **Onshore Wind:** Modeled with a 100 m hub height using the Vestas V90 2000 turbine profile [16].
   - **Offshore Wind (OSW):** Modeled using the same Vestas V90 2000 turbine profile as the onshore sites, as justified in [16]. To capture the highest yield, the two sites with the greatest offshore wind potential in both the Luzon and Visayas grids were selected for these specific simulations.

   Two factors are combined to construct the grid-wide composite profile.

a. *Technical potential weights:* Within each grid, individual Competitive Renewable Energy Zones (CREZ) are assigned a proportional weight for either solar or wind generation. This weighting factor is derived by dividing a specific zone's installed-capacity potential by the sum of the installed-capacity potentials across all zones for that particular energy source.

b. *Hourly normalization*: To determine the hourly composite capacity factor for a specific energy source within a given grid, the individual hourly capacity factors of all constituent zones are multiplied by their respective weights and summed together. This resulting unitless composite factor is subsequently multiplied by the

Power Development Plan (PDP) installed capacity target to calculate the total hourly generation in gigawatts (GW). All associated data processing and modeling are implemented in MATLAB.

B. Model Building

1. *Net load profile:* The net load profile is the foundational component of this study. It dictates system behavior for each weather year based on the planning targets set by the PDP. The hourly net load for a given weather-year is computed by deducting the total hourly variable renewable energy generation from the target year's scaled demand profile. The total renewable generation is the aggregate sum of the hourly composite capacity factors for solar, wind, and offshore wind, each scaled by their respective PDP 2023–2050 installed capacity targets.

2. *Recurrence matrix and heatmap development:* In representing the net-load variability, the recurrences for 40 weather-years of the net load events will be enumerated for each amplitude and duration. This follows the algorithm described below to develop this matrix.

   a. *Event counting:* Continuous net-load periods are identified, and the modified counting scheme of Göransson [21] is applied: each event is counted at its maximum duration and at every discrete duration between zero and that maximum.

   The matrix axes and color scale are defined as follows: *x-axis*: amplitude in 1 MW increments; *y-axis:* duration in 1-hour increments; and *recurrence count:* capped at 100 occurrences [5]; and linear scale on all axes, with a zoomed-in view at the first 144 hours.

   The result is a 2D heatmap where the x-axis is amplitude in GW, the y-axis is duration in hours, and the color encodes the recurrence frequency across the 40 weather-years.

C. Simulation and Analysis

To assess the sufficiency of the generation capacity in PDP 2023-2050, the net load duration and variability curves are tested against the dispatchable generation and storage targets. The analysis comprises a dependable capacity test and a visualization step.

1. *Dependable capacity test:* This test will determine the sufficiency of capacity targets to cover the drops in renewable generation. To perform this analysis, the values in the x-axis (Amplitude in GW) will be subtracted from the dependable capacity target. If the result is less than zero, then there is a capacity gap during those extreme events.

2. *Visualization:* To visualize the results, specific areas are determined and drawn over the net load heatmap: *Capacity Risk Zone:* The right of the vertical axis will be called Capacity Risk Zone where the PDP lacks in dependable capacities; and *Safe Zone:* The area between the duration curve and the vertical axis are the events that the dependable capacity targets can handle easily.

III. Results and Discussion

This section presents a comparative analysis of the simulation results, structured by the three major Philippine power grids: Luzon, Visayas, and Mindanao. For each grid, the analysis evaluates two primary frameworks. First, the Reference Scenario serves as the baseline to assess the standard development trajectory of the energy landscape [4]. Second, Clean Energy Scenario 1 is evaluated to determine the impact of integrating a low penetration of offshore wind into the capacity mix. A detailed, grid-by-grid discussion of these findings is provided below.

A. Luzon

Figures 1a–1f present the Luzon grid's Net Load Recurrence Heatmaps under the Reference Scenario (REF) and Clean Energy Scenario 1 (CES 1). While 2030 projections remain safely within dependable capacity limits, subsequent decades reveal escalating capacity shortfalls.

By 2040, where the target of 50% VRE penetration in the capacity mix, worst case deficits reach 6.8 GW (REF) and 8.2 GW (CES 1). These shortfalls worsen significantly by 2050, widening to 15.6 GW and 17.9 GW, respectively.

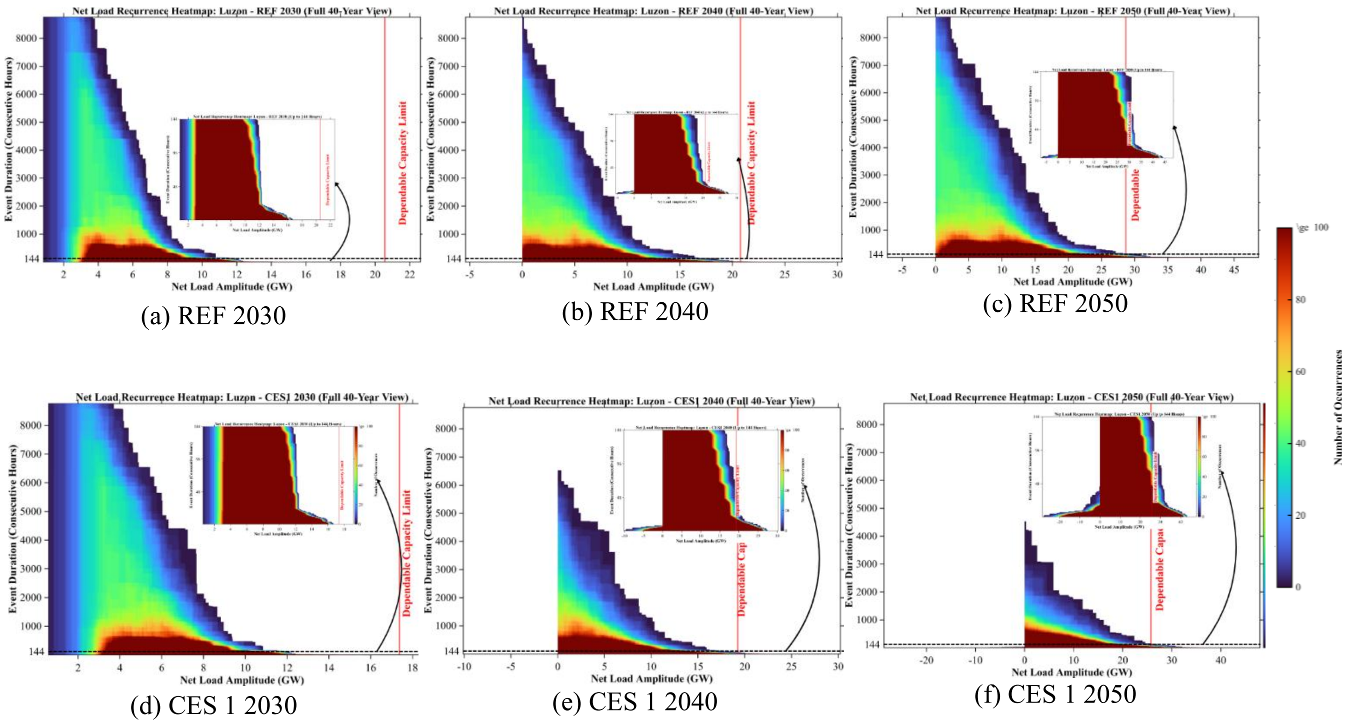
Beyond the magnitude of these peak deficits, the duration of these deficits poses a severe threat to grid reliability. Both scenarios project maximum consecutive shortfalls lasting up to 144 hours (six days), alongside highly recurrent 1–2 GW deficits lasting up to 48 hours. Because these multi-day energy gaps far exceed the capabilities of standard 4-hour Battery Energy Storage Systems (BESS), they highlight a critical risk of extended grid outage.

B. Visayas

Figures 2a-2f present the Visayas grid's Net Load Recurrence Heatmaps (2030–2050) under the Reference Scenario (REF) and Clean Energy Scenario 1 (CES 1).

Under the REF scenario, the net load remains safely within dependable capacity limits throughout the target years. Under CES 1, however, the 2030 reserve margin shrinks to a critically thin 18.47 MW, eventually breaching dependable capacity with a 759 MW shortfall by 2040. However, this deficit is ultimately resolved by 2050, resulting in a 2 GW capacity surplus.

Furthermore, it is observed that as Variable Renewable Energy (VRE) penetration increases across both scenarios, the heatmaps flatten vertically and spread horizontally. This indicates that while the range of net load amplitudes expands, the duration of consecutive net load events significantly decreases. Because of the Visayas grid's lower overall demand, any projected deficits and high-net-load events remain much shorter in duration than the prolonged shortfalls observed in larger systems like Luzon.



(a) REF 2030 (b) REF 2040 (c) REF 2050

(d) CES 1 2030 (e) CES 1 2040 (f) CES 1 2050

Figure 1: Luzon Heatmaps for REF and CES 1

C. Mindanao

Figures 3a-3f present the Visayas grid's Net Load Recurrence Heatmaps (2030–2050) under the Reference Scenario (REF) and Clean Energy Scenario 1 (CES 1).

Under the REF scenario, the net load remains safely within dependable capacity limits throughout the target years. Under CES 1, however, the 2030 reserve margin shrinks to a critically thin 18.47 MW, eventually breaching dependable capacity with a 759 MW shortfall by 2040. However, this deficit is ultimately resolved by 2050, resulting in a 2 GW capacity surplus.

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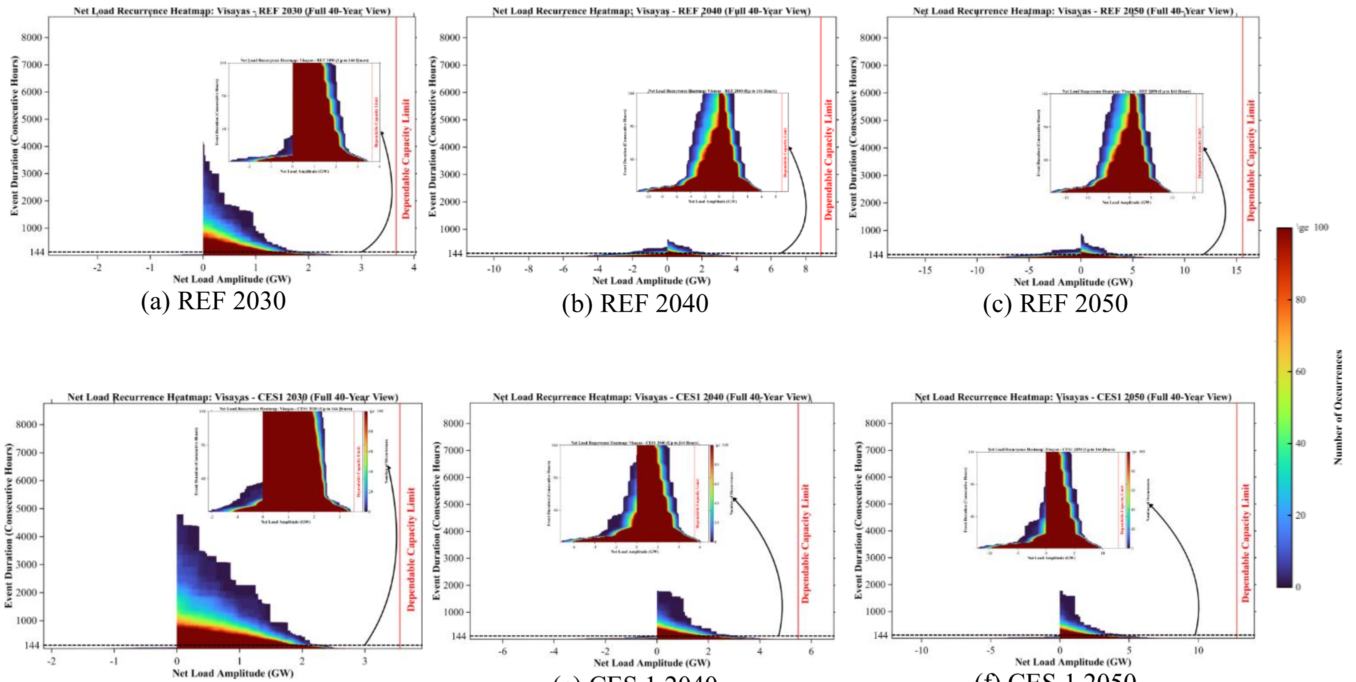
D. Worst Case Net Load Scenarios

Table 1 summarizes the projected gap between dependable capacity and the worst-case net load across the three major grids under both the Reference Scenario (REF) and Clean Energy Scenario 1 (CES 1). A positive value indicates a reliable capacity reserve, while a negative value signifies a supply deficit.

		Dependable Capacity less Worst-Case Net Load (GW)		
Scenario	Year	Luzon	Visayas	Mindanao
Reference Scenario	2025	+6.06 (+47.51%)	+0.96 (+37.62%)	+0.78 (+30.70%)
	2030	+3.86 (+23.09%)	+0.12 (+3.37%)	+0.10 (+3.82%)
	2040	-6.88 (-24.88%)	+2.64 (+42.30%)	-2.05 (-34.28%)
	2050	-15.58 (-35.21%)	+5.57 (+55.57%)	-5.56 (-58.54%)
CES 1	2025	+3.60 (+28.21%)	+0.96 (+37.67%)	+0.55 (+21.54%)
	2030	+0.72 (+4.35%)	+0.002 (0.52%)	-0.01 (-3.88%)
	2040	-8.24 (-30.63%)	-0.76 (-12.14%)	-2.17 (-36.34%)
	2050	-17.88 (-40.99%)	+2.70 (+26.85%)	-4.84 (-50.84%)

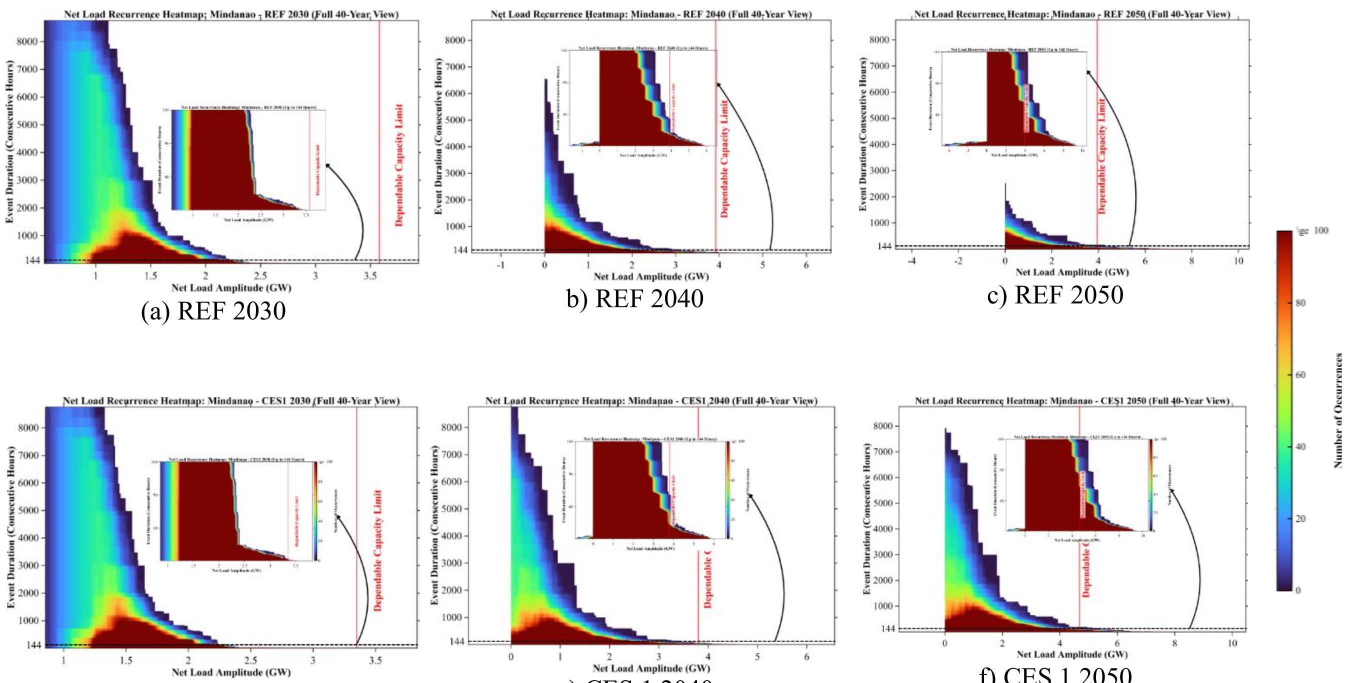
Table 1: Dependable Capacity less Worst-Case Net Load (GW) of Reference Scenario and CES 1 Capacity Outlook for the Three Major Grids: Luzon, Visayas, Mindanao



(a) REF 2030 (b) REF 2040 (c) REF 2050

(d) CES 1 2030 (e) CES 1 2040 (f) CES 1 2050

Figure 2: Visayas Heatmaps for REF and CES 1



(a) REF 2030 b) REF 2040 c) REF 2050

d) CES 1 2030 e) CES 1 2040 f) CES 1 2050

Figure 3: Mindanao Heatmaps for REF and CES 1

IV. Conclusion and Future Work

In this paper, the long-term resource adequacy of the Philippine power system was evaluated against the high penetration targets of the Philippine Development Plan 2023-2050. By utilizing 40 years of historical meteorological data, multi-decadal net load recurrence matrices were generated for the country's three major power grids: Luzon, Visayas, and Mindanao. This study captured the effects of highly variable environmental parameters on future capacity mixes, allowing for the stress testing of the 2025, 2030, 2040, and 2050 expansion scenarios.

The analysis demonstrates a critical risk of prolonged capacity deficiencies as the grid develops toward the target years. Specifically, findings reveal that the Luzon and Mindanao grids risk severe, multi-day capacity shortfalls at 50% renewable penetration by 2040. Under worst-case conditions, these supply deficits could peak at 17.88 GW in Luzon and 5.56 GW in Mindanao.

Finally, this paper recommends including a hydropower curve to further analyze the support of this resource on the grid. It is further recommended to investigate the feasibility of integrating Long-Duration Energy Storage (LDES) technologies. Lastly, future studies can explore the impact of expanded inter-grid connectivity to determine if sharing reserve capacity across regional boundaries can effectively mitigate the extreme, localized net load amplitudes observed in isolated grid simulations.

References

[1] National Renewable Energy Laboratory. “Resource Adequacy”. Accessed: 2024. [Online]. Available: https://www.nrel.gov/research/resource-adequacy.

[2] P. Denholm, "Fundamentals of Resource Adequacy for Modern Power Systems," National Renewable Energy Laboratory (NREL), Golden, CO, NREL Rep. 95306, 2025. [Online]. Available: https://docs.nrel.gov/docs/fy25osti/95306.pdf

[3] H. Shaker, H. Zareipour, and D. Wood, "Impacts of large-scale wind and solar power integration on California's net electrical load," Renewable and Sustainable Energy Reviews, vol. 58, pp. 761–774, 2016, doi: 10.1016/j.rser.2015.12.287.

[4] G. Tziolis, A. Livera, S. Theocharides, J. Lopez-Lorente, G. Makrides, and G. E. Georghiou, "Net load forecasting: A comprehensive literature review," Sustainable Energy Technologies and Assessments, vol. 82, p. 104450, 2025, doi: 10.1016/j.seta.2025.104450.

[5] J. Ullmark, L. G¨oransson, and F. Johnsson, “Representing net load variability in electricity system capacity expansion models accounting for challenging weather-years”, Energy, vol. 316, p. 134 346, Jan. 2025. DOI: 10.1016/j.energy.2024.134346.

[6] Department of Energy, “Power Development Plan (PDP) 2023–2050”, Republic of the Philippines, Government Report, 2023. [Online]. Available: https://legacy.doe.gov.ph/sites/default/files/pdf/electric_power/development_plans/Power%20Development%20Plan%202023-2050.pdf.

[7] International Renewable Energy Agency, “Planning for the Renewable Future: Long-term Modelling and Tools to Expand Variable Renewable Power in Emerging Economies”, IRENA, Abu Dhabi, Tech. Rep., 2017. [Online]. Available: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2017/IRENA_Planning_for_the_Renewable_Future_2017.pdf.

[8] R. Spencer, J. Macknick, A. Aznar, A. Warren, and M. Reese, “Floating Photovoltaic Systems: Assessing the Technical Potential of Photovoltaic Systems on Man-Made Water Bodies in the United States”, National Renewable Energy Laboratory (NREL), Tech. Rep. NREL/TP-6A20-72414, 2019. [Online]. Available: https://docs.nrel.gov/docs/fy19osti/72414.pdf.

[9] H. C. Bloomfield, D. J. Brayshaw, L. C. Shaffrey, P. J. Coker, and H. E. Thornton, “Quantifying the increasing sensitivity of power systems to climate variability”, Environmental Research Letters, vol. 11, no. 12, p. 124 025, 2016. DOI: 10.1088/1748-9326/11/12/124025.

[10] O. Ruhnau and S. Qvist, “Storage requirements in a 100% renewable electricity system: Extreme events and inter-annual variability”, Environmental Research Letters, vol. 17, no. 4, p. 044 018, 2022. DOI: 10.1088/ 1748-9326/ac4dc8.

[11] D. Stenclik and R. Deyoe, “New Resource Adequacy Criteria for the Energy Transition: Modernizing Reliability Requirements”, Energy Systems Integration Group (ESIG), Tech. Rep., 2024. [Online]. Available: https://www.esig.energy/wp-content/uploads/2024/03/ESIG-New-Criteria-Resource-Adequacy-ES-2024.pdf.

[12] R. Gelaro et al., “The Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2)”, Journal of Climate, vol. 30, no. 14, pp. 5419–5454, May 2017. DOI: 10.1175/jcli-d-16-0758.1.

[13] S. Pfenninger and I. Staffell. “Renewables.ninja”. Accessed: 2025. [Online]. Available: https://www.renewables.ninja.

[14] S. Pfenninger and I. Staffell, “Long-term patterns of European PV output using 30 years of validated hourly reanalysis and satellite data”, Energy, vol. 114, pp. 1251–1265, Nov. 2016. DOI: 10.1016/j.energy.2016.08.060.

[15] I.Staffell and S. Pfenninger, “Using bias-corrected reanalysis to simulate current and future wind power output”, Energy, vol. 114, pp. 1224–1239, Nov. 2016. DOI: 10.1016/j.energy.2016.08.068.

[16] N. Lee, A. Dyreson, D. Hurlbut, M. I. McCan, E. Neri, et al., “Ready for Renewables: Grid Planning and Competitive Renewable Energy Zones (CREZ) in the Philippines”, National Renewable Energy Laboratory (NREL), Technical Report NREL/TP-6A20-76235, Sep. 2020. [Online]. Available: https://docs.nrel.gov/docs/fy20osti/76235.pdf.

[17] National Renewable Energy Laboratory. “Renewable Energy Data Explorer”. U.S. Agency for International Development (USAID) and NREL. Accessed: Feb. 2025. [Online]. Available: https://www.re-explorer.org.

[18] A. Frazier, W. Cole, P. Denholm, S. Machen, N. Gates, and N. Blair, “Storage Futures Study: Economic Potential of Diurnal Storage in the U.S. Power Sector”, National Renewable Energy Laboratory (NREL), Technical Report, May 2021. DOI: 10.2172/1785688.

[19] National Grid Corporation of the Philippines, "Operations," National Grid Corporation of the Philippines, 2026. [Online]. Available: https://www.ngcp.ph/operations.

[20] National Renewable Energy Laboratory. “Renewable Energy Data Explorer”. U.S. Agency for International Development (USAID) and NREL. Accessed: Feb. 2025. [Online]. Available: https://www.re-explorer.org.

[21] L. Goransson, “Balancing electricity supply and demand in a Carbon-Neutral northern Europe”, Energies,vol. 16, no. 8, p. 3548, Apr. 2023. DOI: 10.3390/en16083548.